

Benjamin Mark Lewis

Neuro-Data Scientist | Multi-Omic Discovery

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Professional Experience

Actio Biosciences – San Diego, CA

Senior Scientist - Bioinformatics, Research and Development

January 2026 – Present

Bioinformatics Scientist II, Research and Development

April 2024 – December 2025

- Promoted to Senior Scientist in January 2026 in recognition of scientific leadership and cross-program impact.
- Partnered with biology, translational, and executive stakeholders to guide data-driven portfolio decisions
- Led multi-omic biomarker discovery and novel target identification across therapeutic programs.
- Drove target identification efforts resulting in multiple assets advancing to nomination and portfolio prioritization.
- Scientific lead for single-cell and bulk transcriptomic strategy across multiple therapeutic programs.
- Designed and deployed production-grade RNA-seq pipeline (Nextflow, Python, AWS), reducing turnaround time from 2 weeks to <48 hours and standardizing analytics across R&D.

Scientist I, Research and Development – Aspen Neuroscience – San Diego, CA

March 2022 – March 2024

- Cross-functional single-cell and multiomic discovery supporting pipeline and IND-enabling programs.
- Single-cell transcriptomics/multiomics: project design and data analysis lead.
- Functional genomics for pipeline development: project design and data analysis lead.
- Contributed to preclinical and IND-enabling studies to support regulatory filings and clinical progression.

Research and Development Intern, Hardy Diagnostics, Santa Maria, CA

June 2013–December 2013

- Developed chromogenic diagnostic media formulations and authored protocols for 510(k) validation studies.

Education

Ph.D. in Molecular Biology

2022

University of California, San Diego, CA.

- Co-advisors: Dr. Tony Hunter and Dr. Gene Yeo.
- Thesis project focus: Bioinformatic analysis of RNA-binding proteins and mitochondrial biology.
- Micro-MBA program

M.S. in Biology, Specialization: Stem Cell Research and Technology

2015

B.S. in Biochemistry and Microbiology (Double Major)

2013

California Polytechnic State University, San Luis Obispo, CA

Key Qualifications and Skills

- Multiomic data analysis – Single cell multiomics | Transcriptomics, Proteomics, ATAC-seq, CLIP-seq.
- Biomarker assay analysis experience across Metabolon, SomaScan, Olink, and NULISA platforms.
- Bioinformatics and statistical analysis | Predictive machine learning model design and implementation.
- Coding skills – Python, pandas, matplotlib/seaborn, scikit-learn, scanpy/adata, scvi-tools, PyTorch.
 - Linux, AWS, Cloud/High-performance computing environments, GitHub, R.
 - IBM Data Science Professional Certificate | IBM AI Engineering Professional Certificate.
- Molecular Biology – Viral vector design/production | Gene cloning and DNA assembly techniques.
- Gene expression analysis – NGS/qPCR| RNA manipulation and analysis techniques | Immunoblotting .
- Cell Culture – CRISPR/Cas9 gene-editing in iPSCs & cell lines | Stem cell-based disease modeling.
- Cellular Analysis – Extracellular flux analysis | Flow cytometry techniques and analysis (FACS) | Live Cell imaging.
- Immunofluorescence microscopy – Confocal, Brightfield, Plate scanner.

Selected Publications

“LARP4 Is an RNA-Binding Protein That Binds Nuclear-Encoded Mitochondrial mRNAs To Promote Mitochondrial Function”

- **Journal:** RNA, November 2023 | **Authors:** Benjamin M. Lewis, Chae Yun Cho, Hsuan-Lin Her, Orel Mizrahi, Gene W. Yeo, Tony R. Hunter.

“Autologous and Heterologous Cell Therapy for Hemophilia B toward Functional Restoration of Factor IX”

- **Journal:** Cell Reports, May 2018 | **Authors:** S. Ramaswamy, T. Menon, Benjamin M. Lewis, ..., Inder M. Verma.

“Functional Gene Correction for Cystic Fibrosis in Lung Epithelial Cells Generated from Patient iPSC”

- **Journal:** Cell Reports, September 2015 | **Authors:** Amy L. Firth, Tushar Menon, GS. Parker, SJ. Qualls, Benjamin M. Lewis, ..., Fred H. Gage, Inder M. Verma.

Research Training

Ph.D. Student, UCSD Biology department	Summer 2015-March 2022
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Laboratory of Tony Hunter (Co-Advisor) – Salk Institute, La Jolla, CA

Laboratory of Gene Yeo (Co-advisor) - University of California, San Diego, CA

- Ph.D. Project Focus: Post-transcriptional regulation of nuclear encoded mitochondrial mRNAs
- Analysis of CLIP data identified several RNA-binding proteins that bind nuclear encoded mitochondrial mRNAs
- Bioinformatic analysis to show LARP4 binds mRNAs encoding for OXPHOS and mito-ribosomal proteins
- Through quantitative proteomics showed that LARP4 depletion triggers reduction in protein levels of targets
- Functional cellular assays to demonstrate LARP4 contributes to mitochondrial function and homeostasis
- Developed methods to study localized translation at the surface of the mitochondria

CIRM Bridges to Stem Cell Research Intern

July 2014 – March 2015

Laboratory of Inder M. Verma - Salk Institute, La Jolla, CA

Mentor and supervisor: Dr. Tushar Menon, Postdoctoral Researcher

- Completed several CRISPR genome-editing projects in iPSC cell models and in-vivo models
- Generated novel CRISPR-corrected hemophilia B patient derived iPSC lines: Published in Cell Reports
- Designed and constructed the several HDR targeting vectors the in-vitro/in-vivo correction of hemophilia B

Undergraduate Student Research Assistant

January 2012–June 2013

Laboratory of Pat Fidopiastis- Cal Poly, San Luis Obispo, CA

Mentor and supervisor: Dr. Pat Fidopiastis, Professor

- Developed and performed agar plate based mucinase assay screens on a library of *V. fischeri* transposon mutants
- Performed bioinformatic screens secreted enzymes with potential activity on mucin molecules

Posters

“Proximity-labeling Based Methods for Studying Localized mRNA and Translation at the surface of the mitochondria”

- **Lewis BM**
- Poster presented at the 2018 UCSD Biology Department retreat, Lake Arrowhead, USA

“Genome Editing of Stem Cells to Brewing Yeast”

- **Lewis BM**, Menon T, Black M, and Verma IM
- Poster presented at the 2015 CIRM Bridges Meeting, San Francisco, USA

“Genomic Editing of Stem Cells for Modeling and Therapy of Monogenic Diseases”

- Menon T, Firth AL, Parker GS, Gilmore WB, **Lewis BM**, Qualls SJ, Scripture-Adams D, Zack JA, and Verma IM
- Poster presented at the 2014 Meeting on the Mesa, The Salk Institute, La Jolla, USA

Teaching Experience

Discussion Section Instructor - “Our Energy Future”	Spring 2019
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Discussion Section Instructor - “Computational Neurobiology”	Spring 2018
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Lab Instructor - “Biochemical Techniques”	Fall 2016
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University of California, San Diego, CA

Lab Instructor - “Introduction to Cell and Molecular Biology”	Fall 2013
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California Polytechnic State University, San Luis Obispo, CA

Volunteer Service

Science Fair Judge	2019, 2021
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Greater San Diego Science and Engineering Fair

- Volunteer judge for middle school (grades 7–8) biology science fair projects

Science Fair Mentor

2015

San Luis Obispo Elementary Science Fair Mentor

- Volunteer mentor for 4th grade elementary student.